

Intermediate elective option 2

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1. Set up

Packages

```
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrapping axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# adapting Scherer's figure
library(colorspace) # darken() and lighten() for after_scale()
library(glue)       # multi-line label strings
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                   sheet = "Aquatic Insects")
```

```
# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                  sheet = "sites")

# water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling
↪ Data-2026-03-10.xlsx"),
                          sheet = "Water Quality",
                          na = c("", "n/a", "N/A", "over", "173+"))
```

2. Cleaning

sites object

```
# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")
```

Cleaning taxon_list

```
# creating new clean object from taxon_list
taxon_list_clean <- taxon_list |>
  # converting taxon name to snake case (no spaces or capital letters,
  # only underscores)
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))
```

Cleaning water_quality

```
# creating new clean object from water quality
water_quality_clean <- water_quality |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites in ncos_sites
```

```

semi_join(ncos_sites, by = "site") |>
# create new column to join with later data frames
unite("date_site", date, site, remove = TRUE) |>
# group by date_site column
group_by(date_site) |>
# calculate median pH, salinity, DO
summarize(med_ph = median(p_h, na.rm = TRUE),
          med_sal = median(salinity_ppt, na.rm = TRUE),
          med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
# ungroup data frame
ungroup() |>
# filter out salinity outliers
filter(date_site != "2022-08-12_NVBR")

```

Cleaning aq_ins

```

# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
# clean column names
clean_names() |>
# filtering join: only include sites occurring in ncos_sites
semi_join(ncos_sites, by = c("site")) |>
# renaming columns
rename(date = date_on_vial) |>
# select columns of interest
select(date, sample_type, site,
       ostracod,
       copepod,
       hemiptera_corixidae_boatman,
       cladocera,
       nematode,
       diptera_ceratopogonidae,
       annelida_oligochaete,
       diptera_chironomid,
       annelida_polychaete) |>
# filter to only include "filtered beaker" samples
filter(sample_type %in% c("FB 250", "FB250")) |>
# convert the data frame to long format
pivot_longer(cols = ostracod:annelida_polychaete,
             names_to = "taxon",

```

```

      values_to = "abundance") |>
# group by date, site, taxon
group_by(date, site, taxon) |>
# calculate average abundance per liter (sum abundance divided by 7.5
  ↪ liters)
summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
# ungroup data frame
ungroup() |>
# create a new column called `date_site` to join with water quality
unite("date_site", date, site, remove = FALSE) |>
# join with water quality
left_join(water_quality_clean, by = c("date_site")) |>
# join with taxon list
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
# add full names of sites
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "South of Venoco Bridge",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~ "West Pond",
  site == "NDC" ~ "Devereux Creek"
)) |>
# setting factor levels for site
mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm
  ↪ = TRUE),
      site_full = fct_rev(site_full))

```

Problem 1. Explain your inspiration

Which visualization you chose for your inspiration

- I chose Cédric Scherer's Bill Dimensions of Brush-tailed Penguins (TidyTuesday 2020/31). Panel A of his figure is a scatter plot of bill length vs. bill depth coloured by penguin species. There are two signature techniques I am copying: two-layer `geom_point`, which is a filled layer for the body of each point and a transparent layer with a white outline on top, and per-species text labels placed at fixed positions showing the species name in bold colour with a lightened multi-line stats block immediately below.

Why it makes sense for your dataset of choice.

- The cleaned aquatic invertebrate data has the structure Panel A was built for. There are two continuous variables per observation, median salinity at the survey and log-transformed abundance per L. It is a small number of groups to color by, and a continuous variable to map onto bubble size. The labelled blocks let each group carry its own median statistics inside the plot rather than in a side legend.

Which variables from your dataset will be shown in your visualization, and what visual components they map onto (axes, shapes, colors, etc.)

- x-axis: med_sal median salinity
- y-axis: log_ave_lit = $\log(\text{ave_lit} + 1)$
- point colour and label colour: taxon_group , Microcrustaceans, Aquatic insects and Worms, which are 3 taxa each
- point size: ave_lit, average abundance per L
- group name + median salinity, median log abundance: per-group label block

Problem 2. Plan your figure

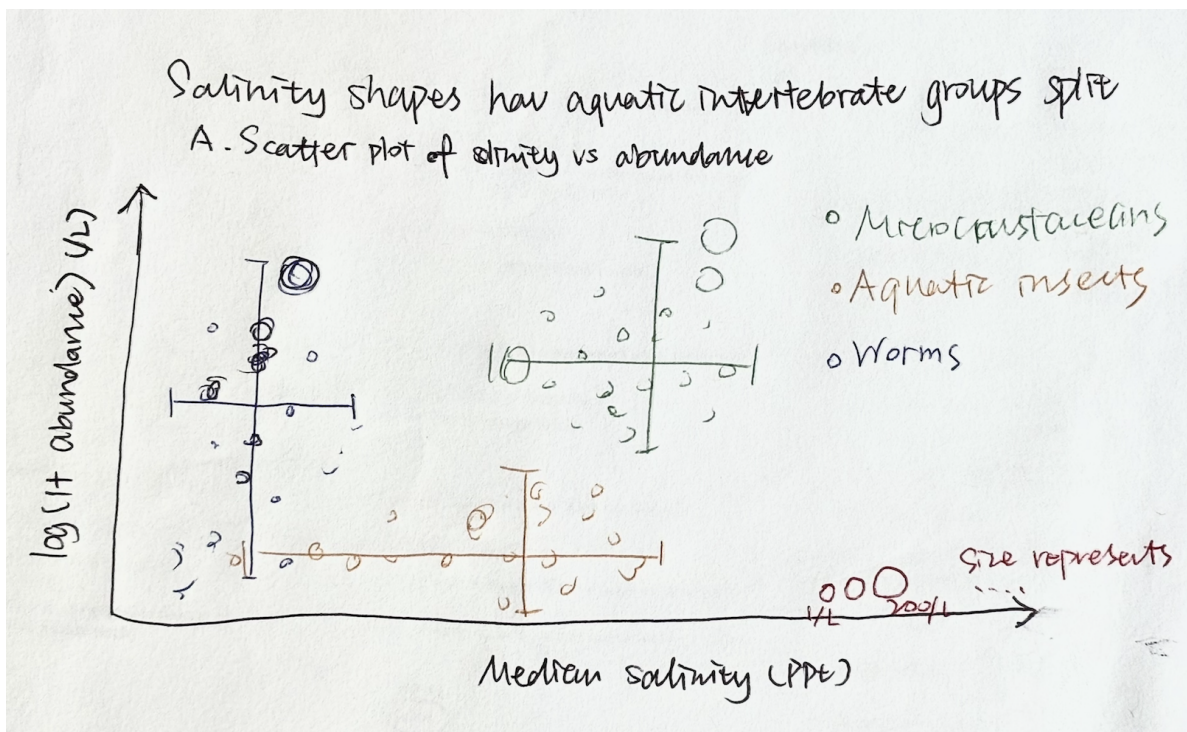


Figure 1: Planning sketch of the figure.

Problem 3. Code your figure

3a. Wrangle

```
# collapse the 9 surveyed taxa into 3 ecological groups
group_lookup <- tribble(
  ~taxon,                ~taxon_group,
  "ostracod",            "Microcrustaceans",
  "copepod",             "Microcrustaceans",
  "cladocera",           "Microcrustaceans",
  "hemiptera_corixidae_boatman", "Aquatic insects",
  "diptera_ceratopogonidae",    "Aquatic insects",
  "diptera_chironomid",         "Aquatic insects",
  "annelida_oligochaete",       "Worms",
  "annelida_polychaete",        "Worms",
  "nematode",                   "Worms"
)

# build the data frame the figure will read from
fig_data <- aq_ins_clean |>
  # attach the 3-group ecological label
  left_join(group_lookup, by = "taxon") |>
  # focus on presences (abundance > 0)
  filter(ave_lit > 0) |>
  # drop surveys that have no matching water quality reading
  filter(!is.na(med_sal)) |>
  # log + 1 transform abundance so the scatter spreads out
  mutate(log_ave_lit = log(ave_lit + 1),
    # factor order: micro - insects - worms
    taxon_group = factor(taxon_group,
      levels = c("Microcrustaceans",
        "Aquatic insects",
        "Worms")))

```

3b. Code the figure

```
# parallel of Scherer's df_peng_summary, with hand-picked label positions
df_inv_summary <-
  # hand-picked (x, y) label positions for each taxon group

```

```

tribble(
  ~taxon_group, ~x, ~y,
  "Microcrustaceans", 38, 6.4,
  "Aquatic insects", 55, 3.2,
  "Worms", 55, 5.6
) |>
# attach per-group summary stats by joining on taxon_group
full_join(
  fig_data |>
    # group at the taxon-group level
    group_by(taxon_group) |>
    # count observations and compute medians for each axis variable
    summarize(n = n(),
              med_sal_median = median(med_sal, na.rm = TRUE),
              log_ave_lit_median = median(log_ave_lit, na.rm = TRUE)),
  by = "taxon_group"
) |>
# build the multi-line label string shown next to each group on the plot
mutate(label = glue("Median salinity: {format(round(med_sal_median, 1),
  ↪  nsmall = 1)} ppt\nMedian log abund: {format(round(log_ave_lit_median,
  ↪  2), nsmall = 2)}\nn = {n}"))

```

```

# three saturated colors, same as (Scherer's pal)
pal <- c("Microcrustaceans" = "#159090",
        "Aquatic insects" = "#FF8C00",
        "Worms" = "#A034F0")

# theme adapted from Scherer's theme_update, without Neutraface fonts
theme_set(theme_minimal(base_size = 12, base_family = "serif"))
theme_update(
  panel.grid.major = element_line(color = "grey92", size = .4),
  panel.grid.minor = element_blank(),
  axis.title.x = element_text(color = "grey30", margin = margin(t = 7)),
  axis.title.y = element_text(color = "grey30", margin = margin(r = 7)),
  axis.text = element_text(color = "grey50"),
  axis.tick = element_line(color = "grey92", size = .4),
  legend.position = "top",
  plot.title = element_text(hjust = 0, color = "black",
                            size = 21, margin = margin(t = 10, b = 35)),
  plot.subtitle = element_text(hjust = 0, face = "italic", color = "grey30",
                               size = 14, margin = margin(0, 0, 25, 0)),
  plot.title.position = "plot",

```

```

plot.caption = element_text(color = "grey50", size = 9, hjust = 0,
                             lineheight = 1.05,
                             margin = margin(30, 0, 0, 0)),

plot.caption.position = "plot",
plot.margin = margin(rep(20, 4))
)

```

```

scat <-
# base plot: salinity on x, log abundance on y
ggplot(fig_data, aes(med_sal, log_ave_lit)) +
# filled point layer with alpha (Scherer's first geom_point)
geom_point(
  # fill color by taxon group, bubble size by abundance per L
  aes(fill = taxon_group, size = ave_lit),
  # filled circle with separate border
  shape = 21,
  # hide the border on this layer
  color = "transparent",
  # semi-transparent fill
  alpha = .3
) +
# white-outlined transparent point layer (Scherer's second geom_point)
geom_point(
  # same size mapping as the fill layer
  aes(size = ave_lit),
  # same shape
  shape = 21,
  # white outline ring
  color = "white",
  # no fill so the layer below shows through
  fill = "transparent"
) +
# bold group name at the hand-picked label position
geom_text(
  # one row per taxon group
  data = df_inv_summary,
  # label coordinates from the tribble; colour matches the bubbles
  aes(x = x, y = y, label = taxon_group, color = taxon_group),
  # serif font (Scherer used Neutraface)
  family = "serif",
  # bold weight
  fontface = "bold",

```

```

    # text size
    size = 5.0
) +
# lightened median-stats block just below the label
geom_text(
  data = df_inv_summary,
  # placed 0.35 units below the bold name; lighten the group colour
  aes(x = x, y = y - .35, label = label, color = taxon_group,
      color = after_scale(lighten(color, .3))),
  family = "serif",
  # smaller than the bold name
  size = 3.2,
  # tighter line spacing
  lineheight = .9,
  # adjust at top so text grows downward
  vjust = 1
) +
# "bubble size" annotation, placed at the bottom next to the size legend
annotate("text",
  # bottom area, right
  x = 50, y = 0.7,
  label = "Bubble size represents average abundance per litre",
  # match the rest of the figure
  family = "serif",
  # italic for the explanatory note
  fontface = "italic",
  # matches the legend bubble outline
  color = "pink4",
  # small caption-style text
  size = 3,
  # tighter line spacing
  lineheight = .9,
  # adjust
  hjust = 0, vjust = 0.5
) +
# allow text/elements to extend beyond the panel
coord_cartesian(clip = "off") +
# x range covers full salinity span, tick every 20 ppt
scale_x_continuous(limits = c(-2, 80),
  breaks = seq(0, 80, by = 20),
  expand = c(0, 0)) +
# y range covers log abundance span, tick every 2 log-units
scale_y_continuous(limits = c(-0.3, 8),

```

```

        breaks = seq(0, 8, by = 2),
        expand = c(0, 0)) +
# custom palette, no colour legend
scale_color_manual(values = pal, guide = "none") +
# same palette for fills
scale_fill_manual(values = pal, guide = "none") +
# bubble-size legend
scale_size(
  # no legend title
  name = "",
  # three reference sizes
  breaks = c(1, 50, 200),
  # labels for the three breaks
  labels = c("1/L", "50/L", "200/L"),
  # min and max point radii
  range = c(1.5, 14)
) +
# labels under each bubble; outline colour matches the annotation
guides(size = guide_legend(label.position = "bottom",
  override.aes = list(color = "pink4",
    stroke = .8,
    fill = NA))) +

theme(
  # bottom area, right of the annotation
  legend.position = c(.78, .08),
  # bubbles laid out in a row
  legend.direction = "horizontal",
  # narrow keys
  legend.key.width = unit(.01, "lines"),
  # small grey serif text
  legend.text = element_text(size = 8, family = "serif", color = "grey50")
) +
labs(
  # x-axis label
  x = "Median salinity (ppt)",
  # y-axis label
  y = "Log(1 + average abundance per L)",
  title = "Salinity shapes how aquatic invertebrate groups split",
  subtitle = "Scatterplot of salinity vs log abundance",
  # multi-line caption
  caption = paste(
    "Data: ENV5 193DD aquatic invertebrate surveys (NCOS, UCSB).",
    "Visualization adapted from Cedric Scherer's",

```

```

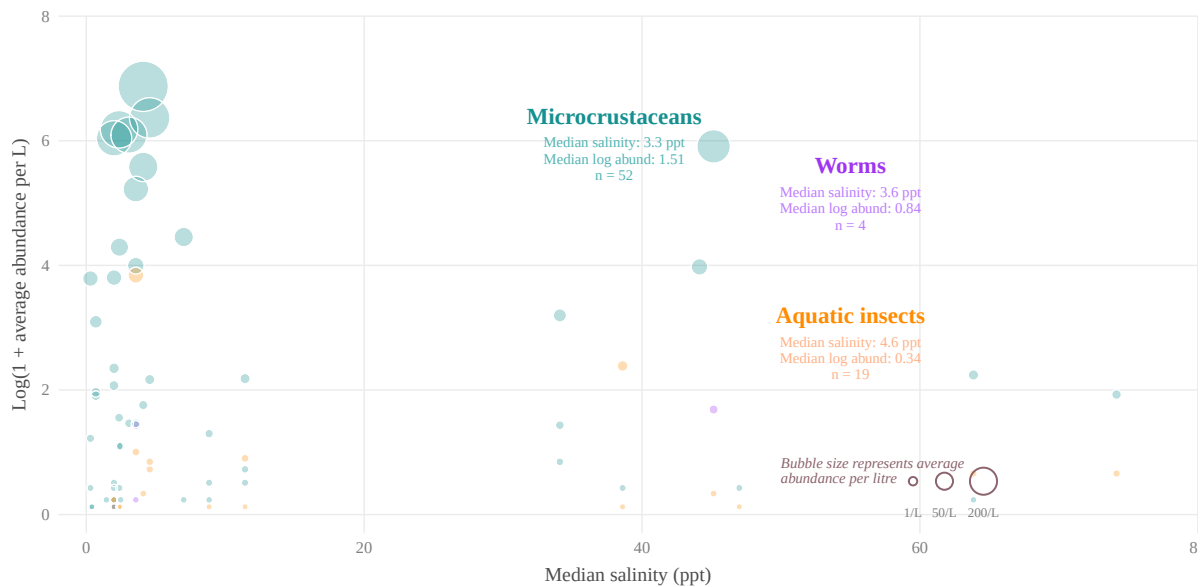
    "'Bill Dimensions of Brush-tailed Penguins'.",
    sep = " "
  )
)

# display
scat

```

Salinity shapes how aquatic invertebrate groups split

Scatterplot of salinity vs log abundance



Data: ENV5 193DD aquatic invertebrate surveys (NCOS, UCSB). Visualization adapted from Cedric Scherer's 'Bill Dimensions of Brush-tailed Penguins'.

3c. Save figure

```

# save to code folder so the README can display it
ggsave(filename = here("code", "final_figure.png"),
  plot      = scat,
  width     = 10,
  height    = 7,
  dpi       = 300,
  bg        = "white")

```

Problem 4. Describe your process

Process of using Scherer's code to create a visualization

- I copied the 2020_31_PalmerPenguins.Rmd chunk-by-chunk and swapped his variables for mine.
- His `df_peng_summary`, which is a `tribble()` of hand-picked label positions joined to a per-species `summarize()` and pasted into a `glue::glue()` label, became my `df_inv_summary` with `taxon_group` in place of `species` and `med_sal/log_ave_lit` in place of `bill_length_mm`.
- His scatter chunk transferred almost no change: both `geom_point` layers, both `geom_text` layers (bold name, then 'after_scale, and the coord-cartesian scales.

Challenges encountered as made visualization

- Scherer's figure uses three Neutraface display fonts that don't work with default, so I change to `family = "serif"` and lost a little of the original's characters. The bubble-size mapping required hand-tuning because abundances per L run from <1 to >200, while Scherer's body-mass range is much tighter and his default `scale_size` worked out of the box.
- I also had to hand-pick the three (x, y) label positions inside the `tribble()` so the per-group label blocks didn't overlap the data clusters. Also, the location of annotation need adjust many times by hand.
- I started by also copying Scherer's two cross-shaped `geom_errorbar`, which is the median SD in both x and y, but the resulting error bars looked busy and overlap with each other on top of the bubble points and I removed them.

How visualization differs from visualizations you made in class, for assignments, or for a previous elective

- Previous class and assignment figures were single-`ggplot` calls with default themes and no in-plot annotation. This figure carries multiple annotation layers, which are two `geom_point` layers per group and two `geom_text` blocks per group. They composed into a single labelled scatter with a custom theme, bubble-size legend, and derived colours. It is the first figure I have built where group identity and group statistics are shown inside the plot rather than in a side legend or marginal panel.
- I learned the new package `glue` for multi-line label strings. And package `colorspace` for `darken()` and `lighten()` for `after_scale()`.